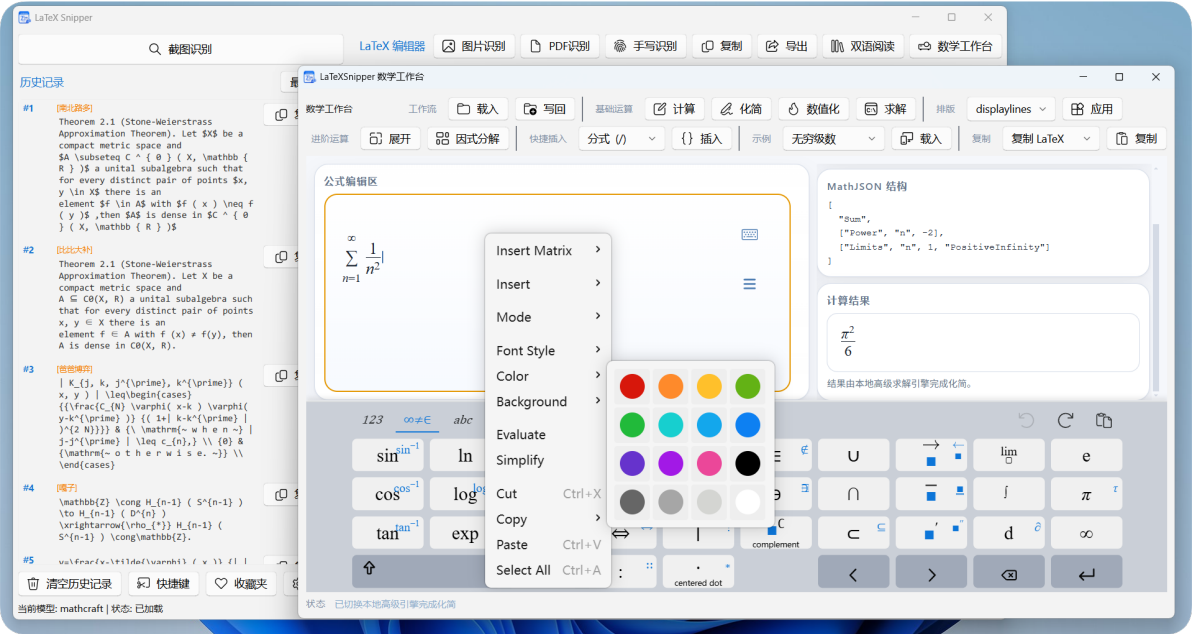




LaTeXSnipper



v2.3.2_Final_Stable |



LaTeXSnipper

-
-
- [MathCraft ONNX](#)
-
-
-
-
- [PDF](#)
-
- [Linux / macOS](#)
-
-

- [Bug](#)

📁 • Office 📁📁📁

- [📁](#)



- [Word](#) 📁
- [PowerPoint](#) 📁
- [📁](#)
- [📁](#)
- [📁](#)

📁 • MathCraft 📁📁📁

- [MathCraft OCR](#) 📁
- [📁](#)

📁 • LaTeXSnipper 📁📁

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📁📁📁📁📁📁📁📁

LaTeXSnipper 📁 — 📁MathCraft OCR📁PDF/Pandoc 📁

📁📁📁📁📁📁📁📁📁📁

📁MathCraft 📁PDF 📁Pandoc 📁

📁

📁Python 📁apt/brew 📁

📁

- Windows📁
- Linux📁python3 -m venv 📁Debian/Ubuntu 📁python3-venv📁
- macOS📁Python 3.10+📁Homebrew Python 📁python.org 📁

#

- **PDF** ☐ **PDF** ☐ **Markdown** ☐ **LaTeX** ☐
- ☐ ☐ ☐ ☐
- ☐ ☐ **PDF** ☐ **PDF** ☐
- ☐ ☐ ☐ **MathLive** ☐
- ☐ / ☐ **LaTeX** ☐ **Markdown** ☐ **MathML** ☐ **HTML** ☐ **OMML** ☐ **SVG** ☐ **Pandoc** ☐
- ☐ / ☐ ☐ ☐

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☐ ☐

- ☐ ☐ **MathCraft OCR** ☐ ☐ ☐ ☐ + ☐ ☐ ☐ ☐
- ☐ ☐ **OpenAI-compatible** ☐ **Ollama** ☐ **MinerU** ☐ **GLM-OCR** ☐ **PaddleOCR-VL** ☐ **Qwen2.5/**
Qwen3-VL ☐ **Ollama Vision** ☐ **MinerU Native** ☐

☐ ☐ **PDF** ☐ **PPI** ☐ ☐ ☐ ☐ **OpenAI-compatible / Ollama PDF** ☐

LaTeX ☐ **MathCraft** ☐ **MathCraft** ☐

PDF ☐ ☐ ☐

PDF ☐ ☐

- ☐ ☐ **MathCraft** **PDF** ☐
- ☐ ☐ ☐
- **MinerU** ☐ **Markdown** ☐ **Markdown** ☐ **LaTeX** ☐
- ☐ **PDF** ☐
- **PDF** ☐ **PPI** ☐ **90-300** ☐ **150 DPI** ☐ **200 DPI** ☐
- **PDF** ☐ **Markdown** ☐ **Assets** ☐

☐ ☐ ☐ ☐

☐ ☐ **PDF OCR** ☐

- **PDF** ☐ **PDF** ☐
- **PyMuPDF** ☐ **PDF** ☐ **PDF** ☐ **OCR** ☐
- **PDF** ☐ ☐ **Poppler** ☐ **Fitz** ☐ **Poppler** ☐ **Fitz** ☐

- [Argos Translate](#) [Azure Translator](#) [Google Cloud Translation](#) [DeepL API Free](#)
- Argos [🔗](#)
- [Azure](#) [Google](#) [DeepL](#) [🔗](#)

📄📄📄📄**MathCraft ONNX**📄📄📄📄

MathCraft ONNX 📄📄📄

`warmup()` `warmup` 📄

`python -m mathcraft_ocr models check` 📄

📄📄📄📄📄📄📄 / 📄📄 / 📄📄📄

📄📄 📄"model xxx downloading"📄

📄📄

- [GitHub Releases](#) 📄
- [🔗](#)
- `urllib` 📄

📄📄

- [GitHub](#)
- `HTTP_PROXY / HTTPS_PROXY` 📄
-
- [MathCraft](#) 📄

-Windows📄%APPDATA%\MathCraft\models\ -Linux/macOS📄/.mathcraft/models/

📄📄📄📄

`python -m mathcraft_ocr models check` 📄

`python -m mathcraft_ocr warmup --profile mixed --provider auto` 📄

onnxruntime 問題発生

エラー: Failed to import onnxruntime [missing get_available_providers]



onnxruntime

- pip
- CPU [GPU]

解決

- pip uninstall onnxruntime onnxruntime-gpu
- [GPU] pip install onnxruntime-gpu
- [CPU] pip install onnxruntime

CUDA / GPU 問題

エラー: warmup [CUDA] cuda wasn't able to be loaded [cublas]

解決

- onnxruntime-gpu [CUDA / cuDNN]
- CUDA [onnxruntime-gpu]
-
- [GPU]

解決

- 環境変数 **GPU** [MATHCRAFT_FORCE_ORT_CPU=1 [CPU]
- 環境変数 **GPU** [CUDA Toolkit [cuDNN [onnxruntime-gpu [onnxruntime]
-

インストールディレクトリ / ディレクトリ

エラー: warmup [invalid protobuf] failed to load model [sha256 mismatch]

解決

解決

- Windows [%APPDATA%\MathCraft\models\<model_id>\] linux/macOS [~/mathcraft/models/<model_id>/]
-

- `python -m mathcraft_ocr models check`

ONNX

warmup CUDA out of memory

GPU

- GPU
- `MATHCRAFT_FORCE_ORT_CPU=1` CPU

API

```
_validate_config() provider mineru model_name base_url  
model_name
```

API → Base URL

OCR 50

vLLM / Ollama / ONNX 7B+ 50

- A curl Ollama API
- B 120-180
- C Ollama `ollama run <model_name>`

Ollama 部署 Base URL 设置

地址 127.0.0.1:11434"

配置



http://localhost:11434/v1 Ollama 地址 /v1

- https:// http:// Ollama HTTPS
- Ollama
- Ollama llama serve

配置

http://127.0.0.1:11434/api/tags JSON

部署地址 / 部署

地址 xxx"

地址

地址

- Ollama llama list
- OpenAI-compatible v1/models

API 部署 API Key

地址 401 "

地址 DeepSeek OpenAI API Key Ollama

地址 API Key

配置

API Key

LaTeXSnipper API Key Base URL

MinerU 部署部署部署部署

地址 404 409

MinerU healthfile_parse

- MinerU http://127.0.0.1:8000/health

>

- MinerU 409 MinerU

OpenAI-compatible Ollama

v1/models 404

Ollama api/tagsv1/models

"OpenAI-compatible" "Ollama"

- Ollama → **Ollama**
- DeepSeek / OpenAI / Groq → **OpenAI-compatible**
- MinerU → **MinerU**
- MathCraft ONNX →

LaTeX

LaTeX"

Python

Python

📄

- 📄 **Windows** 📄📄📄📄 PythonPython 3.11 📄
- **Linux/macOS** 📄📄📄📄 Python 3.10+📄
- >📄📄📄/📄📄📄 Python 3.11📄Windows 📄

```
#Python 3.11 📄📄📄📄  
python3.11 -m venv .venv  
  
#Windows 📄  
.\.venv\Scripts\activate  
  
#Linux/macOS 📄  
source .venv/bin/activate  
  
#Windows 📄📄📄📄  
pip install -r requirements.txt  
  
#Linux 📄📄📄📄📄 Linux 📄📄📄  
pip install -r requirements-linux.txt  
  
#macOS 📄📄📄📄  
pip install -r requirements-macos.txt
```

PyInstaller 📄📄📄📄📄📄📄

📄📄 📄GitHub Release 📄exe 📄git clone 📄

📄📄 📄Windows 📄Python 3.11 📄Linux/macOS 📄PyInstaller 📄Python 📄

📄📄

- 📄📄📄📄 📄Release 📄
- 📄📄📄📄📄 Windows 📄requirements.txt📄Linux/macOS 📄requirements-linux.txt / requirements-macos.txt📄requirements-build.txt

Linux/macOS 📄📄📄📄📄📄

Linux/macOS 📄tools/deps/python311 📄MathCraft📄pandoc 📄Python 📄

```
~/\.latexsnipper/deps/python311
```

📄📄📄📄📄

Linux/macOS `$usr/lib/latexsnipper/app`

`$/.latexsnipper/deps/python311/venv/pyenv.cfg`

Linux `.deb` `python3` `python3-venv` macOS `.deb` `python3`

#Homebrew

`brew install python`

#`python.org` `macOS`

<https://www.python.org/downloads/macos/>

pip `pywin32` / `PyQt6`

`pip install -r requirements.txt`

Ensure `pip` `setuptools` / `wheel` Python Windows `python-3.11.0-amd64.exe` Linux/macOS `Python 3.10+`

- `pywin32` `wheel`
- `PyQt6`
- `PyPI`

- `PyPI`

```
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple -r requirements.txt
```

- Linux/macOS `requirements-linux.txt` `requirements-macos.txt`
- `pywin32` Windows Windows `pip` `PyPI` `Python` `wheel`

Program Files /

`C:\Program Files\`

`%APPDATA%\latexsnipper\JAC`

- GitHub `%LOCALAPPDATA%\LaTeXSnipper`
- `C:\Program Files\`

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📁 📁

📁 Windows 📁 APPDATA% 📁 ASCII 📁 📁 ONNX 📁

>

📁 📁 MATHCRAFT_HOME 📁 ASCII 📁 C:\mathcraft_data📁

📁📁📁📁📁📁

Windows 📁📁📁📁📁📁

📁📁 Ollama / MinerU 📁 LaTeXSnipper 📁

📁📁 Windows Defender 📁

📁📁

- 📁 Python / Ollama 📁
- 📁
- 📁 Ollama 📁 0.0.0.0 📁 127.0.0.1

📁📁📁📁📁

📁📁 📁

📁📁 📁 api.github.com📁

📁📁

- 📁
- 📁 GitHub Releases 📁
- 📁 %USERPROFILE%\latexsnipper/logs/ 📁

📁📁📁📁 / 📁📁

📁📁 exe 📁

📁📁 PyInstaller 📁 exe 📁

📁📁

- 📁

- [GitHub Releases](#) [SignPath](#) [Release](#)



PDF

PDF

- DPI 100 200

- PDF

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- PDF 140-170 DPI

- 200-300 DPI

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- PDF PDF Markdown/LaTeX PDF MinerU Markdown

LaTeX Markdown

PDF MinerU

PDF PDF

/

4K

Base64 3%

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環境構築

インストール



インストール

- インストール環境
- インストール **Linux/macOS** Linux Wayland macOS
- インストール Windows Linux macOS Docker

インストールPandoc

インストール Word / EPUB / Typst

インストール PandocPandoc

インストール

- "PANDOC"
- Pandoc<https://pandoc.org>PATH
- LaTeX / Markdown / MathML / HTML Pandoc

インストール / インストール

インストール

インストール

- OCR MathCraft
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-

インストール

- MathCraft
-
-

如何安装使用各种数学软件/编辑器

如何安装使用 MathCraft



- 如何安装使用 MathCraft
- OCR Markdown + LaTeX

如何安装使用各种数学软件/编辑器

如何安装使用 PDF

如何安装使用

- XeLaTeX ctexart
- amsmath amssymb mathtools geometry
- PDF
- 如何安装使用

如何安装使用各种数学软件/编辑器

如何安装使用

如何安装使用

- 如何安装使用
- Compute Engine

如何安装使用

- $\mathbb{R}^n \times \mathbb{R}^m$
- SymPy/mpmath
- 如何安装使用

如何安装使用各种数学软件/编辑器

如何安装使用

如何安装使用

- 如何安装使用
- 如何安装使用

Pandoc 使用法

ファイル docx epub

Pandoc UTF-8



- Pandoc 3.0
- LaTeX PDF ctex
- Word Word

PDF 使用法

PDF

- PDF OCR
-
- Argos API Key

- PDF OCR PDF
- Argos Translate Argos
- AzureGoogle DeepL

Linux / macOS

LaTeXSnipper WindowsLinuxmacOS PDF

- Windows Win32 Linux pynput 11 Wayland macOS Carbon
- Ctrl+Ctrl+Shift+
- Ctrl+F
- Windows Qt Linux Qtgrimptime-screenshot portal macOS Qt screenshot

- 安装 / 卸载 Windows [Linux](#) [macOS](#) [Docker](#)
- 安装 Windows [Linux](#) [Wayland](#) [macOS](#) [Carbon](#)
- 安装 Windows [GitHub](#) [Store](#) [CPU](#) [Linux/macOS](#) [/.latexsnipper/deps/python311](#) [Python 3.10+](#)
- 安装 Windows [Inno](#) [Store](#) [MSIX](#) [GitHub Release](#) [Linux](#) [Debian/Ubuntu](#) [.deb](#) [macOS](#) [dmg](#) [app.zip](#)

安装

[Ctrl+C](#) [Ctrl+Shift+C](#)

[Linux](#) [Wayland](#) [Wayland](#)

[macOS](#) [Carbon](#) [Linux](#)

Linux: Wayland 安装

安装 [Wayland](#)

安装 [Wayland](#) [Qt](#)

安装

- [grim + slurp](#) [wlroots](#) [gnome-screenshot](#) [GNOME](#)
- [Qt](#)
- [Qt](#)

Linux 安装

[Python](#) [grim](#) [aim](#) [gnome-screenshot](#)

[Qt](#) [Qt](#)

Linux: 安装 Wayland 安装

安装 [EGL not available](#) [No physical devices](#) [GLzone not found](#) [Failed to get system egl display](#)

安装 [MathCraft GPU](#) [Qt WebEngine / Chromium](#) [Wayland](#) [WSL](#) [DRI render](#) [EGL/GPU](#)

安装 [LaTeXSnipper](#) [Linux](#) [K11](#) [GPU](#) [/dev/dri/renderD*](#)

安装

```
#Linux
LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS=1 latexsnipper

#Linux Qt/GPU
LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS=1 latexsnipper
```

Wayland/XCB/QtWebEngine Debian/Ubuntu

```
sudo apt install xwayland libxcb-cursor0 libxcb-icccm4 libxcb-image0
libxcb-keysyms1 libxcb-render-util0 libxcb-xinerama0 libxkbcommon-x11-0
```

macOS:

macOS

Apple notarization

- → →
- `attr -cr /Applications/LaTeXSnipper.app`

macOS:

- → → LaTeXSnipper
-

macOS:

LaTeXSnipper macOS Carbon Pynput

macOS: pip pip

pip / setuptools / wheel

macOS Python Python 3.10+/.latexsnipper/deps/python311Python
venv / ensurepip / pip

- `[H]omebrew Python` `[b]rew install python`
- `[p]ython.org` `[m]acOS Python`
- `> [l]aTeXSnippet`

```

#### & ###
Windows: %USERPROFILE%\\.latexsnipper
Linux:    ~/.latexsnipper/
macOS:    ~/.latexsnipper/

Linux/macOS Python #####
Linux:    ~/.latexsnipper/deps/python311
macOS:    ~/.latexsnipper/deps/python311

#####MathCraft ONNX##
Windows: %APPDATA%\MathCraft\models
Linux:    ~/.mathcraft/models/
macOS:    ~/.mathcraft/models/

```

Python 入門

```
PATH = Path('Python')
```

- where pythonWindows which pythonLinux/macOS Python
-
-

100

hook

LaTeXSnippet



Bug

🐛

"""""" "" "" **issue** ""

≡≡

Bug

🐛

1. 🐛

%USERPROFILE%\latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnippet\logs\

🐛 **logs** 🐛

\$tderr

2. 🐛

crash-native.log

3. 🐛

exe Python

4. 🐛

5. 🐛

6. 🐛

🐛

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• **GitHub Issues** <https://github.com/SakuraMathcraft/LaTeXSnippet/issues>

- **GitHub Discussions** <https://github.com/SakuraMathcraft/LaTeXSnippet/discussions>



README Quick Start

README

- **Windows** `python -m venv .venv → pip install -r requirements.txt → python src/main.py`
- **Linux** `python3 -m venv .venv → pip install -r requirements-linux.txt → python src/main.py`
- **macOS** `python3 -m venv .venv → pip install -r requirements-macos.txt → python src/main.py`
- requirements-linux.txt `requirements-macos.txt` `requirements.txt` `Linux` `Pyinput` `macOS`

`python311/` `tools/deps/python311/` `Python`

- python311/ `Windows` `Python` `ruff/pyright/pytest`
- tools/deps/python311/ `IDE` `Actions` `requirements*.txt` `ruff` `pytest` `pyright`
- Linux/macOS `tools/deps/python311/` `./latexsnipper/deps/python311`

Windows

```
py -3.11 -m venv tools\deps\python311
.\tools\deps\python311\python.exe -m pip install --upgrade pip wheel setuptools
.\tools\deps\python311\python.exe -m pip install -r requirements.txt
.\tools\deps\python311\python.exe -m pip install -r requirements-build.txt
.\tools\deps\python311\python.exe -m pip install ruff pytest pyright
```

□□□□

- `python311/` — Windows □
- `tools/deps/python311/` — □
- > • `~/.latexsnipper/deps/python311` — Linux/macOS □

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`docs/developer_code_standards.md` □

```
. \tools\deps\python311\python.exe -m ruff check .  
. \tools\deps\python311\python.exe -m pytest test  
. \tools\deps\python311\python.exe -m pyright  
. \tools\deps\python311\python.exe -m compileall -q src mathcraft_ocr test
```

□□□□

`ruff` `pytest` `pyright` `requirements*.txt` □

□

```
. \tools\deps\python311\python.exe -m pip install ruff pytest pyright
```

pyrightconfig.json □□□□

`tools/deps/python311` `python311/` `Windows` `pyright` □

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`tools/deps/python311` `python311` □

インストールガイド

Office 環境

LaTeXSnipper Office Windows VSTO Word PowerPoint Ribbon LaTeXSnipper Bridge LaTeXSnipper LaTeX

インストール

LaTeX LaTeX → OMML / PNG OCR LaTeXSnipper Office

前提条件

- **Windows** Windows 10 22H2 Windows 11
- **Office** Microsoft 365 Office 2024 / 2021 / 2019 Office LTSC 2024 / 2021
- .NET Framework 4.8 Microsoft Edge WebView2 Runtime Windows 11 M365
- **Office** 32 64 Office macOS Office
- **LaTeXSnipper** Office 1.27.0.0.1:28765 Bridge

インストール

インストール

- Word PowerPoint Office DLL
-
- HKLM

安装



1

- Word PowerPoint
-
- HKLM x2 x64 Office
- VSTO
- Office

安装

VSTO

Windows" → → "Unins000.exe"

-
- HKLM
- VSTO
- Office

安装

Inno Setup

```
#  
OfficePluginSetup-2.3.2.exe /silent  
  
#  
OfficePluginSetup-2.3.2.exe /verysilent  
  
#  
OfficePluginSetup-2.3.2.exe /dir="D:\Tools\LaTeXSnipper"
```


Word 安装



Word **LaTeXSnippet**

- 安装
- 配置
- 使用

OMML 安装

Word **Bridge** **LaTeX** **Word** **OMML** **Office Math Markup Language** **OMML** **Word**

OMML 安装

- OMML **Word**
- **LaTeXSnippet** **LaTeX**
- **Word** **LaTeXSnippet** **LaTeX**



- 安装
- 配置
- 使用

LaTeXSnippet 安装

LaTeXSnippet

- 安装 **LaTeX**
- 配置
- 使用

LaTeXSnippet

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Word □

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- > □□□□□□ □
- □□□□□ □

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OCR□□□□

PowerPoint □□□□

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PowerPoint □LaTeXSnipper □Word □

- □□□□ □
- □□□□ □
- □□□□ □

□□□□□□

PowerPoint □LaTeX □DPI PNG □Shape.Tags□

□ Word □□□

- PowerPoint □OMML
- □—PowerPoint □
- Word □

□□□□□

- □□□□□ □LaTeX □
- □□□□□ □



Alt Text LaTeXSnipper LaTeXSnipper formula"



Word OCR

MathLive

LaTeX



- **MathLive** MathLive
- **LaTeX** LaTeX
- Bridge
- OCR
-

MathLive



- MathLive
- 1.6
- /
- **Esc**

MathLive

- Ctrl+Enter
- Esc

□□□□□□

Word □□□□ Word □PowerPoint □PPT □



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- **Word** □ □
- **PowerPoint** □ □
- □□□ □Ctrl+Enter □Esc □

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□□Word □PowerPoint□

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- Word / PowerPoint □
- □
- □
- □

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□□□ □Word □PowerPoint□LaTeXSnipper □

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1. □Word / PowerPoint → □→ □→ □
2. □□□COM □□□
3. □**LaTeXSnipper**
4. □

5. 

 LaTeXSnippet 

1. 

2. 

3. 

  LaTeXSnippet 

1.  LaTeXSnippet 

2.  Office 

3.  127.0.0.1:28765 

4.  Bridge  LATEXSNIPPER_OFFICE_BRIDGE_URL 

  Microsoft Edge WebView2 Runtime 

1.  <https://go.microsoft.com/fwlink/p/?LinkId=2124703>  WebView2 Runtime

2. Windows 11  Microsoft 365  WebView2

 **OCR**  

  OCR 

  Office 

  **Office** 

  Office 

 Office 

□□□

1. □→□→□→□→□
2. LaTeXSnipper□
3. >Office Resiliency KCU\Software\Microsoft\Office\{App}\Resiliency□

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□□□ Windows"□→□→ LaTeXSnipper.OfficePlugin.WordVstoAddIn □

LaTeXSnipper.OfficePlugin.PowerPointVstoAddIn □

□□□ VSTO ClickOnce □

□□□ · **MathCraft** □□□□□

MathCraft OCR □□□□

MathCraft OCR LaTeXSnipper ONNX Runtime PDF □

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- GitHub<https://github.com/SakuraMathcraft/MathCraft-Models>
- PyPI<https://pypi.org/project/mathcraft-ocr/>

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- □□□□□□□ □
- **ONNX** □□□□ ONNX RuntimeCPU CUDA GPU □
- □□□□□□ PDF □
- □□□□□□□ □

□□□□□

```
pip install "mathcraft-ocr[cpu]"  
# GPU □□□  
pip install "mathcraft-ocr[gpu]"
```

PyPI ONNX Runtime onnxruntime onnxruntime-gpuLaTeXSnipper □

□□□□□□

Abstract Algebra (p.18)

□



CHAPTER 1. PRELIMINARIES

U

PROOF. (1) If $x \in A \cup B$, then the theorem follows immediately since both A and B are empty set. Otherwise, we must show that $(A \cup B) \subseteq A' \cap B'$ and $(A \cup B) \supseteq A' \cap B'$. Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. So, in either case, $x \in A$ or $x \in B$. By the definition of the complement, $x \in A$ implies $x \notin A'$ and $x \in B$ implies $x \notin B'$. Therefore, $x \notin A' \cap B'$ and we have $(A \cup B) \subseteq A' \cap B'$. To show the reverse inclusion, suppose that $x \in A' \cap B'$. Then $x \notin A$ and $x \notin B$. So, $x \in A'$ and $x \in B'$. Thus $x \in A \cup B$ and so $(A' \cap B') \subseteq (A \cup B)$. Hence, $(A \cup B) = A' \cap B'$ and the proof of (2) is left as an exercise.

Example 1.4 Other relations between sets often hold true. For example,

$$(A \setminus B) \cap (B \setminus A) = \emptyset$$

to see that this is true, observe that

$$\begin{aligned} (A \setminus B) \cap (B \setminus A) &= (A \cap B') \cap (B \cap A') \\ &= A \cap A' \cap B \cap B' \\ &= \emptyset. \end{aligned}$$

Cartesian Products and Mappings

Given sets A and B , we can define a new set $A \times B$, called the *Cartesian product* of A and B , consisting of ordered pairs. That is,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 1.5 If $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \{a, b\}$, then $A \times B$ is the set

$$\{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

and

$$A \times C = \{(x, a), (x, b), (y, a), (y, b)\}.$$

we define the *Cartesian product* of $A_1, A_2, \dots, A_n$ as

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i \text{ for } i = 1, \dots, n\}.$$

If $A_1 = A_2 = \dots = A_n$, we often write A^n instead of $A_1 \times A_2 \times \dots \times A_n$ (where A is written n times). For example, the set $\mathbb{R}^3$ consists of all of 3-tuples of real numbers.

Subsets of $A \times B$ are called *relations*. We will define a *mapping* or *function* f from a set A to a set B to be a special type of relation where each element a of A is associated with a unique element b of B such that $(a, b) \in f$. Another way of saying this is that for every element a in A , there is a unique element b in B such that $(a, b) \in f$. We usually write $f(a) = b$ or $A \rightarrow B$ instead of writing down ordered pairs $(a, b) \in A \times B$, we write $f(a) = b$ or $f: A \rightarrow B$. The set A is called the *domain* of f .

$$f(A) = \{f(a) : a \in A\} \subseteq B$$

$f(A)$ is called the *range* or *image* of f . We can think of the elements in the function's domain as input values and the elements in the function's range as output values.

Dynamics Journal (p.5)

□

where

$$\begin{aligned} \hat{F}_{11}(u_n, v_n, \delta) &= -\frac{(1+b)x^*\delta}{2a} - \frac{a \sin x^* u_n^2}{2} - \frac{(1+b)x^*\delta^2}{4a^2} \\ &\quad - \frac{a \cos x^* u_n^3}{6} - \frac{(1+b)x^*\delta^3}{8a^3} + O(\|(u_n, v_n, \delta)\|^4), \\ \hat{F}_{12}(u_n, v_n, \delta) &= -\frac{(-x^*b^2 + 2a \sin x^* - 2x^*b - x^*)\delta}{2a} \\ &\quad + \frac{a(b-1) \sin x^* u_n^2}{2} + \frac{(1+b)^2 x^* \delta^2}{4a^2} - \cos x^* \delta u_n \\ &\quad + \frac{a(b-1) \cos x^* u_n^3}{6} + \frac{(1+b)^2 x^* \delta^3}{8a^3} + \frac{\sin x^* \delta u_n^2}{2} \\ &\quad + O(\|(u_n, v_n, \delta)\|^4). \end{aligned}$$

Consider the following system

$$\begin{pmatrix} u_{n+1} \\ v_{n+1} \\ \delta_{n+1} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} + \begin{pmatrix} \hat{F}_{11}(u_n, v_n, \delta_n) \\ \hat{F}_{12}(u_n, v_n, \delta_n) \\ 0 \end{pmatrix}, \quad (15)$$

where

$$\begin{aligned} a_{11} &= -1 + a \cos x^*, & a_{12} &= 1, \\ a_{13} &= -\frac{(1+b)x^*}{2a}, & a_{21} &= a(1-b) \cos x^*, \\ a_{22} &= -b, & a_{23} &= \frac{-2a \sin x^* + x^*(b+1)^2}{2a}, \end{aligned}$$

$$\begin{aligned} \hat{F}_{11}(u_n, v_n, \delta_n) &= -\frac{a \sin x^*}{2} u_n^2 - \frac{(1+b)x^*\delta_n^2}{4a^2} \\ &\quad - \frac{a \cos x^*}{6} u_n^3 - \frac{(1+b)x^*\delta_n^3}{8a^3} \\ &\quad + O(\|(u_n, v_n, \delta_n)\|^4), \\ \hat{F}_{12}(u_n, v_n, \delta_n) &= \frac{a(b-1) \sin x^*}{2} u_n^2 + \frac{(1+b)^2 x^* \delta_n^2}{4a^2} \\ &\quad - \cos x^* \delta_n u_n + \frac{a(b-1) \cos x^*}{6} u_n^3 \\ &\quad + \frac{(1+b)^2 x^* \delta_n^3}{8a^3} + \frac{\sin x^*}{2} \delta_n u_n^2 \\ &\quad + O(\|(u_n, v_n, \delta_n)\|^4). \end{aligned}$$

We construct an invertible matrix

$$T = \begin{pmatrix} -\frac{\sec x^*}{a} & -\frac{1}{b-1} & \frac{\sin x^*}{2a \cos x^* - 2b - 2} \\ 1 & 1 & e \\ 0 & 0 & 1 \end{pmatrix},$$

where

$$e = \frac{-a(a \sin x^* - (1+b)x^*) \cos x^* + 2a \sin x^* - x^*(1+b)^2}{2a(-b-1+a \cos x^*)},$$

and use the translation

$$\begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} = T \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix}.$$

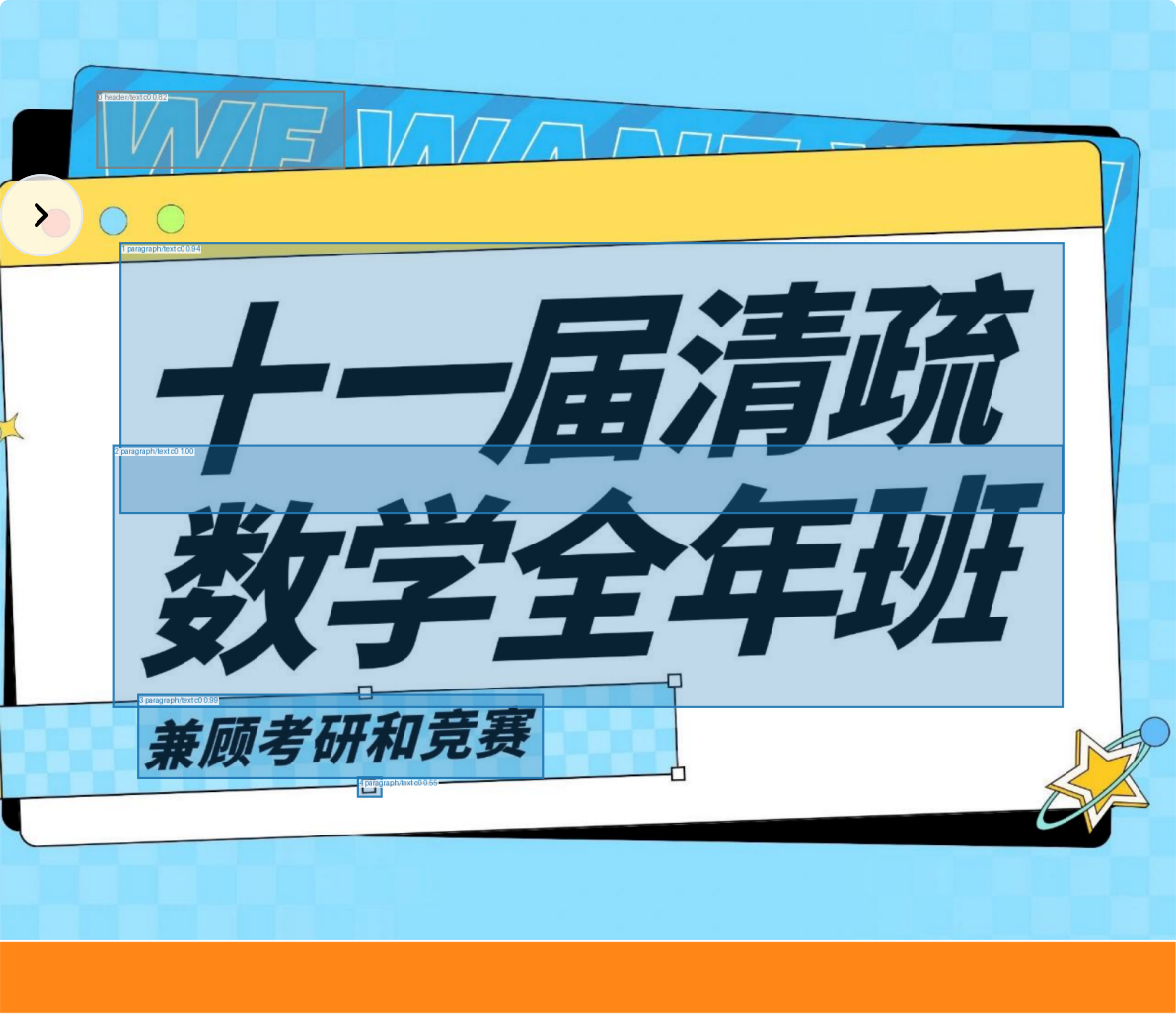
then the system (15) becomes

$$\begin{pmatrix} \tilde{x}_{n+1} \\ \tilde{y}_{n+1} \\ \tilde{\delta}_{n+1} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -b + a \cos x^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix} + \begin{pmatrix} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ 0 \end{pmatrix}, \quad (16)$$

where

$$\begin{aligned} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) &= -\frac{\tan x^* \sec x^* \tilde{x}_n^2}{2a} - \frac{\tan x^* \tilde{x}_n \tilde{y}_n}{b-1} + \frac{\sin x^* \tan x^* \tilde{\delta}_n \tilde{x}_n}{2 \cos x^* (a-2b-2)} \\ &\quad - \frac{a \sin x^* \tilde{y}_n^2}{2(b-1)^2} + \frac{a \sin^2 x^* \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a \cos x^* - b - 1)} \\ &\quad - \frac{\tilde{\delta}_n^2}{8(a \cos x^* - b - 1)^2 a^2} \left((2x^*(b+1)^3 + a^3 \sin^3 x^* \right. \\ &\quad \left. - 4ax^*(b+1)^2 \cos x^* + 2a^2 x^*(b+1) \cos^2 x^*) \right) \\ &\quad + \frac{\sec^2 x^* \tilde{x}_n^3}{6a^2} + \frac{\sec x^* \tilde{x}_n^2 \tilde{y}_n}{2a(b-1)} - \frac{\tan x^* \tilde{x}_n^2 \tilde{\delta}_n}{4a(a \cos x^* - b - 1)} \\ &\quad + \frac{\tilde{x}_n \tilde{y}_n^2}{2(b-1)^2} - \frac{\sin x^* \tilde{x}_n \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a \cos x^* - b - 1)} + \frac{a \cos x^* \tilde{y}_n^3}{6(b-1)^3} \\ &\quad - \frac{a \cos x^* \sin x^* \tilde{y}_n^2 \tilde{\delta}_n}{4(b-1)^2(a \cos x^* - b - 1)} + \frac{a \cos x^* \sin^2 x^* \tilde{y}_n \tilde{\delta}_n^2}{8(b-1)(a \cos x^* - b - 1)^2} \\ &\quad - \frac{\tilde{\delta}_n^3}{48(a \cos x^* - b - 1)^3 a^3} \left(6a^3 x^*(b+1) \cos^3 x^* \right. \\ &\quad \left. - 18a^2 x^*(b+1)^2 \cos^2 x^* - 6x^*(b+1)^4 \right. \\ &\quad \left. + a(a^3 \sin^3 x^* + 18x^*(b+1)^3) \cos x^* \right) \\ &\quad + O(\|(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n)\|^4), \end{aligned}$$

$$\begin{aligned} \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) &= \frac{(b-1) \tan x^* \sec x^* \tilde{x}_n^2}{2a} + \tan x^* \tilde{x}_n \tilde{y}_n \\ &\quad - \frac{\tilde{\delta}_n \tilde{x}_n}{2a(a \cos x^* - b - 1)} (ab \sin x^* \tan x^* - a \cos x^* \\ &\quad + 2b + 2 - a \sec x^*) \\ &\quad + \frac{a \sin x^* \tilde{y}_n^2}{2b-2} - \frac{\tilde{x}_n \tilde{y}_n^2}{2b-2} - \frac{a \cos x^* \tilde{y}_n^3}{6(b-1)^2} \\ &\quad - \frac{\tilde{\delta}_n \tilde{y}_n}{2(b-1)(a \cos x^* - b - 1)} \left(-a \cos^2 x^* + (2b+2) \cos x^* \right. \\ &\quad \left. + a(b \sin^2 x^* - 1) \right) \\ &\quad + \frac{\tilde{\delta}_n^2}{8a^2(a \cos x^* - b - 1)^2} \left((a^3 b \sin^3 x^* + 2x^*(b+1)^4 \right. \\ &\quad \left. + 4(a \sin x^* - x^*(b+1)^2) a(b+1) \cos x^* - a^3 \sin x^* \right. \\ &\quad \left. - 3(a \sin x^* - \frac{2x^*(b+1)^2}{3}) a^2 \cos^2 x^* \right) \\ &\quad + O(\|(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n)\|^4). \end{aligned}$$



第十一届清疏大学生数学竞赛班讲义

非数学类 & 数学类共用

作者：清疏数学

组织：清疏大学

时间：July 7, 2024

版本：2.0

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Limits and Series (p.1)

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Problem Books in Mathematics



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Ovidiu Furdui

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Limits, Series,
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- 1,417
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- 8.53

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MathCraft OCR 

- ONNX Runtime
- PyTorch
- Manifest
- AI
- PCR
-

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`MATHCRAFT_FORCE_ORT_CPU` `MATHCRAFT_HOME`

Windows□□□□□□□

1. Win+R → `sysdm.cpl` → `tab` → `tab`
2. `"C:\Program Files\Internet Explorer\IEXPLORE.EXE" 1`

Windows□□□□□□□□

```
❏ PowerShell ❏❏❏$env:MATHCRAFT_FORCE_ORT_CPU ="1"
```

Linux / macOS□□□□□□

- ~/.bashrc □ ~/.zshrc □□□□□

```
export MATHCRAFT_FORCE_ORT_CPU=1
```

```
□□□□ source ~/.bashrc
```

Linux / macOS□□□□□□□□

```
export MATHCRAFT_FORCE_ORT_CPU=1
```

環境変数

- MATHCRAFT_FORCE_ORT_CPU — ☐ CPU ☐ GPU
- MATHCRAFT_HOME — ☐ MathCraft ☐
- ☒ HTTP_PROXY / HTTPS_PROXY — ☐
- MATHCRAFT_LOG_LEVEL — ☐ DEBUG / INFO / WARNING / ERROR
- LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS — Linux ☐ Qt/WebEngine ☐
- LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS — Linux ☐ Qt/GPU ☐
- LATEXSNIPPER_SHOW_CONSOLE — Windows ☐

LaTeXSnipper FAQ · ☒ 2.3.2_Final_Stable · <https://github.com/SakuraMathcraft/LaTeXSnipper>